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Master2 Smart Aerospace & Autonomous System

DIFFERENTIAL METHODS FOR NOISY SIGNAL DIFFERENTIATION:
TOWARDS A NEW GENERATION OF MODEL-FREE CONTROLLERS

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ABSTRACT

The estimation of derivatives from noisy signals is challenging because exact differentiation inherently amplifies high-frequency components, which include noise. This makes the derivative estimate highly unstable, leading to unreliable and erratic results. This project investigates three numerical differentiation methods: the high-gain observer, differentiation using annihilators, and sliding mode differentiation. The high-gain observer approach used Lyapunov-based stability principles to enhance robustness against measurement disturbances. Differentiation using annihilators, grounded in algebraic estimation techniques, employs integral formulations to extract derivatives. The sliding mode differentiation method, which is based on robust control theory, makes the derivative estimates converge in finite time, while noise is naturally suppressed by discontinuous control mechanisms. The numerical simulations were studied to compare these approaches across various signal conditions, analyzing their performance in terms of accuracy, noise rejection, and computational efficiency.

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ACRONYMS

ADRC	Active Disturbance Rejection Control
ES	Extended State
FIR	Finite Impulse Response
HGC	High Gain Controllable
HGO	High-Gain Observer
HOSMD	High-Order Sliding Mode Differentiator
MFC	Model-Free Control
MHGO	Modified High Gain Observer
SMC	Sliding Mode Control

NOMENCLATURE

ϵ	Small positive constant for stability
γ	Observer gain parameter
κ	Tuning parameter in algebraic differentiation
λ	Sliding mode control parameter
μ	Correction factor for peak suppression
ω	Adaptive gain parameter
Π	Annihilator operator
$\rho(s)$	Weighting function in algebraic differentiation
τ	Time delay for numerical differentiation
ξ	Tuning parameter in algebraic differentiation
A	System matrix
B	Input matrix
C	Output matrix
$f(t)$	Unknown disturbance or uncertainty
$H(t)$	Observer gain matrix
L	Bound on higher-order derivatives
$P(t)$	Gain matrix in Brunovski form
$s(t)$	Sliding surface
T	Sampling period
$u(t)$	Control input
z_i	Internal states in sliding mode differentiation

CHAPTER 1: INTRODUCTION

1.1 What is signal differentiation?

Signal differentiation is the process of computing the derivative of a signal, which is a fundamental operation in control systems, estimation, and signal processing. However, signal differentiation in a noisy environment is difficult. To address this issue, various advanced methods have been developed, including high-gain observers, sliding-mode differentiators, and algebraic differentiation techniques.

1.2 Why signal differentiation?

Signal differentiation is essential in many engineering applications, including real-time control, fault diagnosis, and state estimation. It is important across various disciplines, including human motion analysis for force and moment calculations, thermal systems for heating and cooling rate estimation, target tracking for velocity and acceleration determination for feedback control, and prediction[2]. In control systems, velocity and acceleration estimation is required for feedback mechanisms, particularly in robotics, automotive systems, and aerospace applications.

Signal differentiation is also enhancing system stability and performance by enabling accurate derivative estimation [1, 6]. In discrete-time implementations, high-gain observer-based differentiation effectively approximates continuous-time derivatives while minimizing computational errors [5]. In fault detection, it allows us to extract higher-order derivatives, which is essential, as sudden changes in system dynamics can indicate potential faults or external disturbances [7].

1.3 Effect of noise on signal differentiation

Noise has a significant impact on signal differentiation, often leading to instability and poor estimation accuracy. Traditional numerical differentiation techniques, such as finite differences, are highly sensitive to measurement noise, which can introduce large fluctuations in derivative estimates. High-frequency components of noise tend to be amplified when computing derivatives, making direct numerical differentiation impractical in many real-world applications [2]. In high-gain observer-based differentiation, excessive observer gains may also lead to noise amplification, reducing the practical effectiveness of the method [3]. Additionally, noise-induced

errors can degrade control system performance, leading to undesirable oscillations or delays in response [1]. Noisy measurements lead to inaccurate data, reducing the accuracy, leads to instability and reduce overall performance of the controller [8].

1.4 How to reduce effect of noise?

Several strategies have been proposed to mitigate the impact of noise on signal differentiation:

- **High-Gain Observers (HGOs):** Used Lyapunov stability principles to improve derivative estimation while counteracting noise effects [1]. HGOs are widely used to estimate signal derivatives while suppressing noise effects. By properly tuning the observer gains, it is possible to achieve a balance between fast convergence and noise attenuation. Time-varying high-gain observers have been proposed to adapt to different noise levels dynamically [3, 7].
- **Sliding-Mode Differentiation:** A method based on robust control theory, ensuring finite-time convergence of derivative estimates and intrinsic noise suppression. These differentiators use discontinuous control strategies to reject noise while accurately estimating derivatives, making them particularly effective in systems affected by bounded disturbances and measurement uncertainties [4, 6].
- **Numerical Differentiation with Annihilators:** An alternative approach is to use algebraic differentiation techniques, where signal derivatives are extracted using polynomial fitting and differential operators. This method significantly reduces noise sensitivity compared to traditional numerical differentiation [2].
- **Artificial Delays in Observers:** Recent studies have explored the use of artificial delays in observer design to improve noise robustness. By introducing a delayed estimation mechanism, these observers filter out high-frequency noise while maintaining accurate derivative estimates [7].
- **Low-Pass Filtering:** Filtering techniques, such as Butterworth and Kalman filters, can be incorporated into differentiation schemes to attenuate noise before computing derivatives. This reduces the influence of high-frequency components while preserving the integrity of signal dynamics [1].

1.5 Objectives of this project

The objective of this project is to study the given paper, understand the algorithms, and test them on Matlab for different signals and different noises.

The main algorithms are :

- High gain observers for signal differentiations by S. Ibrir, Y. Chitour
- High order sliding mode differentiators by A. Levant et al.
- Numerical differentiation with annihilators (Algebraic algorithm) by Michel, Fliess and Cédric Join

Based on the list of papers, we studied and analyzed the concepts, implemented High-Gain Observers (HGOs) and High-Order Sliding Mode (HOSM) differentiators in MATLAB, and evaluated their performance. The comparison was conducted in terms of accuracy, noise sensitivity, and stability under two noise conditions and two test signals and was concluded based on the results.

1.6 Organization of the project

- **Chapter 1: Introduction** lays the groundwork by explaining what signal differentiation is, why it's important, and the challenges it presents, particularly with noise. It also outlines the objectives of this project.
- **Chapter 2: Literature Review** explores existing techniques for signal differentiation, including high-gain observers, numerical differentiation method, and high-order sliding mode differentiators (HOSMDs).
- **Chapter 3: Mathematical Approaches** provides a deeper analysis of different differentiation methods, comparing their accuracy, robustness, and real-world applications.
- **Chapter 4: Simulations, Results, and Discussions** puts these techniques to the test through simulations. We analyze their performance and also explore how differentiation can be applied in model-free control strategies.
- **Chapter 5: Conclusion and Recommendation** wraps up the study by summarizing key findings and suggesting directions for future research.

CHAPTER 2: LITERATURE REVIEW

2.1 High-Gain Observers for Signal Differentiation

2.1.1 Theoretical Foundations: High-Gain Observer Design and the Peaking Phenomenon

High-gain observers offer a robust framework for derivative estimation [3], [1]. Their design centers around constructing an observer whose dynamics closely mirror those of the signal of interest. This allows the observer's state variables to asymptotically converge to the signal's derivatives [3]. The convergence is achieved through a high-gain feedback mechanism that amplifies the difference between the observed signal and the observer's estimate. However, this high-gain approach introduces a significant challenge: the "peaking phenomenon." This phenomenon manifests as an initial, transient period where estimation errors exhibit unbounded or large overshoots before converging to zero [3], [1]. This peaking can severely impact the accuracy and stability of the estimation, especially in the presence of noise. The magnitude of the peaking is directly related to the gain parameter; higher gains lead to faster convergence but also more pronounced peaking.

Chitour's work addresses this peaking problem by introducing time-varying high-gain vectors that allow for a trade-off between fast convergence and peak suppression. The dynamics of the time-varying gain are carefully chosen to achieve attenuating the peaking effect and minimizing sensitivity to noise disturbances [3].

2.1.2 Implementation and Testing in MATLAB: Continuous and Discrete-Time Formulations

Ibrir provides both continuous and discrete-time formulations of high-gain differentiators, making them readily adaptable for MATLAB implementation and testing [1]. The continuous-time formulation provides a theoretical framework, while the discrete-time version is essential for practical implementation using sampled data. The paper explores two distinct structures for high-gain differentiators: the observable canonical form and the controllable canonical form [1]. These structures differ in how the system's dynamics are represented and how the gain is calculated.

The choice between these structures impacts the performance trade-offs between peaking, differentiation error, and noise filtering [1]. MATLAB's Control System Toolbox provides extensive functionalities for simulating these systems. The `lsim` function can be used to simulate

the continuous-time systems, while the `dlsim` function is suitable for discrete-time simulations. Furthermore, MATLAB's signal processing toolbox offers tools for generating various types of noise signals (e.g., white Gaussian noise, colored noise) and analyzing the resulting output signals. This allows for a comprehensive evaluation of the differentiators' performance under various noise conditions and signal characteristics.

2.1.3 Comparative Analysis: Strengths and Weaknesses of High-Gain Observers

High-gain observers offer a mathematically rigorous approach to derivative estimation, providing theoretical guarantees on the convergence of the observer's state to the signal's derivatives under appropriate conditions [3]. This makes them attractive for applications requiring a strong theoretical basis for accuracy. However, several limitations need to be considered. The peaking phenomenon, as discussed earlier, can lead to significant transient errors, particularly when dealing with noisy signals [1]. The sensitivity to noise is another crucial issue; high-gain observers amplify noise, potentially leading to inaccurate derivative estimates, especially at high noise levels. Time-varying high-gain approaches such as Chitour's mitigate the peaking problem but require careful parameter tuning to balance convergence speed and peak suppression [3]. The optimal choice of gain parameters often involves a trade-off between reducing the peaking effect and maintaining sufficient gain for accurate estimation.

2.2 High-Order Sliding Mode Differentiators (HOSMDs)

2.2.1 Theoretical Underpinnings: Sliding Mode Control and Exact Differentiation

High-order sliding mode differentiators (HOSMDs) represent a different approach to derivative estimation, leveraging the principles of sliding mode control (SMC) theory [4]. Unlike high-gain observers, which rely on asymptotic convergence, HOSMDs aim for finite-time convergence to a sliding surface in the state space. The dynamics of the system constrained to this sliding surface corresponds to the desired derivatives of the input signal. The sliding mode is enforced by a discontinuous control law, designed to drive the system towards and maintain it on the sliding surface. This characteristic provides inherent robustness to disturbances and uncertainties.

The key advantage of HOSMDs lies in their ability to achieve theoretically exact derivative estimation, up to a certain order, in the absence of noise [4]. This exactness is achieved after a finite-time transient phase, during which the system's trajectory converges to the sliding surface.

The order of the differentiator (denoted as k) determines the highest-order derivative that can be exactly estimated. The accuracy of the estimation is guaranteed only if an upper bound Lipschitz on the absolute value of the $(k+1)$ th derivative of the signal is known [4]. This bound is crucial in designing the control law that maintains the sliding mode.

2.2.2 Accuracy Analysis and Robustness to Noise: Best-Possible Differentiation Accuracy

Levant et al. provide a rigorous analysis of HOSMD accuracy, establishing, for the first time, the best-possible (non-asymptotic) differentiation accuracy [4]. This analysis goes beyond asymptotic analysis, providing concrete bounds on the estimation error in the presence of noise. The analysis is grounded in the Landau-Kolmogorov inequalities, which relate the bounds on different order derivatives of a function [4]. These inequalities provide fundamental limitations on the accuracy achievable by any differentiation method. Levant et al. show that HOSMDs achieve accuracy close to these fundamental limits. The robustness to noise stems from the inherent insensitivity of sliding mode control to matched disturbances, meaning disturbances that affect the system in the same way as the control input [4]. The sliding mode dynamics effectively filter out these disturbances, leading to robust derivative estimates.

2.2.3 Implementation and Discretization: Continuous and Discrete-Time HOSMDs

Levant et al. [4] present several implementations of HOSMDs, including both continuous and discrete-time versions. The continuous-time versions provide the theoretical foundation, while the discrete-time versions are crucial for practical applications using sampled data. The paper discusses discretization issues in detail, emphasizing the impact of discretization on the accuracy and robustness of the differentiator. The authors propose techniques to address these issues, ensuring that the discrete-time implementation retains the essential properties of the continuous-time differentiator.

The implementation involves carefully choosing the discretization method and parameters to minimize the introduced error. The choice of discretization method can significantly affect the performance of the discrete-time HOSMD. Methods like the Euler method are simple to implement but may introduce significant errors, especially for high-order derivatives. More advanced methods, such as higher-order Runge-Kutta methods, can improve accuracy but increase computational complexity.

The authors also address the problem of estimating the equivalent control and its derivatives, which are crucial components in sliding mode control. They propose a robust exact method for

this estimation, further enhancing the overall accuracy and robustness of the HOSMD [4]. Simulating HOSMDs in MATLAB can be done using the Simulink environment. The discontinuous nature of the sliding mode control law necessitates the use of specialized solvers or techniques within Simulink to handle the discontinuities effectively. The choice of solver can significantly impact the accuracy and efficiency of the simulation.

2.2.4 Comparative Analysis: Advantages and Limitations of HOSMDs

HOSMDs offer significant advantages in terms of noise robustness and the potential for exact differentiation [4]. Their ability to achieve exact differentiation in the absence of noise, after a finite-time transient, is a key strength. The robustness to noise is another significant advantage, making them suitable for applications with high noise levels. However, some limitations must be considered. The requirement for a known upper bound on a higher-order derivative is a significant constraint, limiting the applicability of HOSMDs to situations where such a bound can be reliably determined [4]. The discontinuous control law employed in sliding mode control can lead to chattering, a high-frequency switching phenomenon that can damage actuators and introduce unwanted high-frequency noise into the system [4]. Mitigation techniques, such as boundary layer approaches, are often used to reduce chattering, but this can compromise the accuracy and robustness of the differentiator. The choice of parameters in HOSMDs can also be complex, requiring careful tuning to balance accuracy, robustness, and the suppression of chattering.

2.3 Numerical differentiation with annihilators

2.3.1 Theoretical Framework: Annihilators and Iterated Integrals

Mboup, Join, and Fliess present a distinct approach to noisy signal differentiation, rooted in algebraic methods [2]. This approach avoids the state-space representation used in high-gain observers and HOSMDs. The core idea centers around the concept of annihilators. An annihilator is a differential operator that, when applied to the signal, eliminates the smooth component, leaving only the noise. This is achieved by constructing a differential operator whose roots match the signal's dominant frequencies or characteristics [2]. The differentiator then employs iterated integrals of the noisy signal to estimate the derivatives. The use of iterated integrals helps to smooth out the noise and extract the underlying smooth signal. The algorithm is designed to estimate the derivative at a specific point in time, making it suitable for pointwise derivative estimation.

The introduction of delayed estimates is a key feature of this method, significantly enhancing its performance [2]. The delay acts as a smoothing filter, reducing the impact of noise on the derivative estimates. The optimal delay is determined based on the characteristics of the signal and the noise. The choice of delay involves a trade-off; a shorter delay provides a more accurate representation of the instantaneous derivative but is more susceptible to noise, while a longer delay reduces noise sensitivity but may introduce a lag in the estimation. The mathematical framework involves the use of operational calculus and differential algebra, allowing for the derivation of explicit formulas for the derivative estimates in terms of iterated integrals of the noisy signal [2].

2.3.2 Implementation with FIR Filters: Practical Considerations

The algebraic approach translates into a practical implementation using finite impulse response (FIR) digital filters [2]. FIR filters are well-suited for digital signal processing due to their linear phase response and inherent stability. The algorithm's output can be expressed as the convolution of the noisy signal with a specific impulse response. This impulse response is determined by the chosen annihilator and the desired derivative order. The coefficients of the FIR filter are derived directly from the algebraic expressions for the derivative estimates [2].

The use of FIR filters simplifies the implementation in MATLAB, as the filter function provides a straightforward way to apply the filter to the sampled noisy signal. The filter's coefficients are directly calculated from the algebraic formulas derived in the paper, making the implementation relatively simple and efficient. The ability to easily change the filter coefficients allows for flexibility in adjusting the differentiator to different signal and noise characteristics. The frequency response of the FIR filter can be analyzed using MATLAB's frequency analysis tools to understand its filtering characteristics and how it impacts the accuracy of the derivative estimations.

2.3.3 Comparative Analysis: Advantages and Limitations of the Algebraic Approach

The algebraic method offers a unique perspective on noisy signal differentiation, distinguishing itself from the state-space-based approaches of high-gain observers and HOSMDs [2]. This unique perspective allows it to bypass some of the limitations associated with state-space models, such as the peaking phenomenon and the need for a known derivative bound. The implementation using FIR filters simplifies the analysis and testing in MATLAB, making it a computationally efficient approach.

However, the performance of the algebraic method is heavily dependent on the choice of annihilation

lator and the characteristics of the signal being differentiated [2]. The selection of an appropriate annihilator requires careful consideration of the signal's properties and the noise characteristics. An improperly chosen annihilator can lead to inaccurate or unstable derivative estimates. The method's effectiveness is also influenced by the sampling rate and the length of the observation window, requiring careful consideration of these factors during implementation. Furthermore, the method assumes that the underlying signal is smooth enough to be characterized by a differential operator. For signals with sharp discontinuities or high-frequency components, the performance might be affected.

CHAPTER 3: MATHEMATICAL APPROACHES

3.1 High-Gain Observers for Signal Differentiation

The author of this paper addressed the challenging problem of numerically estimating the time derivatives of a bounded signal $y(t)$ without any *a priori* knowledge of its dynamics [1]. He also developed two primary linear structures for time-derivative tracking, and a discrete version was also provided for digital implementation. The methods exploit Lyapunov theory, high-gain techniques, and canonical forms, while explicitly dealing with the *peaking phenomenon* and *noise filtering*.

A time-varying observer is designed in the observable canonical form where the state $x(t)$ is driven by

$$\dot{x}(t) = A x(t) + H^{-1}(t) C^\top (y(t) - C x(t)) \quad (3.1)$$

The matrix $H(t)$ evolves according to a Lyapunov-like equation:

$$\dot{H}(t) = -\omega H(t) - A^\top H(t) - H(t)A + C^\top C.$$

After a transient, each state $x_i(t)$ approximates the $(i - 1)^{\text{th}}$ derivative of $y(t)$, i.e., $x_i(t) \approx y^{(i-1)}(t)$. The choice of the initial matrix $H(0)$ influences the *peaking phenomenon* (large transient overshoots).

Since this method is model based, the author approximated the above equation into: [eq. 2]

$$\begin{aligned} \dot{x}_i(t) &= x_{i+1}(t) + C_n^i \gamma^i (y(t) - x_1(t)), \quad i = 1, \dots, n-1, \\ \dot{x}_n(t) &= C_n^n \gamma^n (y(t) - x_1(t)). \end{aligned} \quad (3.2)$$

where C_n^i are the binomial coefficients

Error Analysis and Noise Filtering was presented by using a Lyapunov function $V(x(t)) = x^\top(t)H(t)x(t)$ to show that

$$\dot{V}(x(t)) \leq -\gamma V(x(t)), \quad (3.3)$$

ensuring exponential decay of the estimation error. Using Young's inequality, an error bound of the form

$$\|x_i(t) - y^{(i-1)}(t)\|_\infty \leq K_i \frac{\|y^{(i)}(t)\|_\infty}{\gamma}, \quad 1 \leq i \leq n,$$

was also derived.

In Controllable Canonical (Brunovski) approach, the signal $y(t)$ appears as a control input in a state-space representation: [eq. 14]

$$\dot{x}(t) = A x(t) - B B^\top P^{-1}(t) (x(t) - C^\top y(t)). \quad (3.4)$$

The gain matrix $P(t)$ is computed from the dual Lyapunov equation:

$$\dot{P}(t) = -\omega P(t) - P(t)A^\top - A P(t) + B B^\top.$$

This form enhances noise filtering and reduces peaking, especially when the differentiation order n is increased. The author approximated the above state-space form for the model-free approach as follows: [eq. 40 in the paper]

$$\begin{aligned} \dot{x}_i(t) &= x_{i+1}(t), \quad 1 \leq i \leq n-1, \\ \dot{x}_n(t) &= -\gamma^n (x_1(t) - y(t)) - \sum_{i=1}^{n-1} C_n^i \gamma^i x_{n-i+1}(t), \end{aligned} \quad (3.5)$$

To minimize the peaking phenomenon, the author replaced γ by $\mu * t$ which is acting like an adaptive gain.

The discrete version of these methods are also given in the paper at equations (45-49).

3.2 High-Order Sliding Mode Differentiators (HOSMDs)

A robust, *sliding-mode (SM) based differentiation* technique with an emphasis on finite-time convergence and optimal accuracy was presented [4]. Key contributions included the design of higher-order sliding mode (HOSM) differentiators that estimate the derivatives of an input signal even in the presence of *bounded noise*, and the extraction of equivalent control from a discontinuous SM control input via nonlinear filtering. The reviewed paper also extended the methods to the case where the parameter L (**bounding the $(k+1)^{\text{th}}$ derivative**) may vary with time, and discretization techniques were developed to implement these differentiators in sampled-data settings.

The approach in [4] relied on driving a sliding variable $s(t)$ to zero in finite time using high-

frequency control switching. This idea is extended to the differentiation problem by designing differentiators that, under ideal conditions (noise-free), yield the exact derivatives after a finite-time transient.

Both recursive and non-recursive forms of HOSM differentiators are presented. The differentiators are homogeneous systems (scaling invariance) ensuring finite-time convergence and robustness. Homogeneity plays a central role in the analysis and design of these differentiators.

The authors used **Landau–Kolmogorov Inequalities** to relate the bounds on different orders of a function's derivatives. In particular, for any function $\varphi(t)$ with a bounded $(k+1)^{\text{th}}$ derivative, the inequalities

$$M_{I,i}(\varphi) \leq \beta_{I,i,k} M_{I,k+1}(\varphi)^{\frac{i}{k+1}} M_{I,0}(\varphi)^{\frac{k+1-i}{k+1}}, \quad i = 0, \dots, k, \quad (3.6)$$

are employed to establish the best-possible differentiation accuracy given by [4]:

$$\max_{f, f_0 \in W^{(k)}(L)} \left| \hat{f}_0^{(i)}(t) - f_0^{(i)}(t) \right| = K_{i,k} (2L)^{\frac{i}{k+1}} \varepsilon^{\frac{k+1-i}{k+1}}, \quad (3.7)$$

with, e.g., $K_{1,1} = \sqrt{2}$ for $k = 1$.

Differentiators with Variable Parameter $L(t)$ also presented to provide additional robustness but necessitates modifications to ensure convergence. Based on homogeneous assumptions, they also provided recursive and non-recursive differentiation as presented in [eq. 5, 6, & 7 in the paper]. The non-recursive equation is:

$$\begin{aligned} \dot{z}_0 &= -\tilde{\lambda}_k L^{\frac{1}{k+1}} [z_0 - f(t)]^{\frac{k}{k+1}} + z_1, \\ \dot{z}_1 &= -\tilde{\lambda}_{k-1} L^{\frac{2}{k+1}} [z_0 - f(t)]^{\frac{k-1}{k+1}} + z_2, \\ &\vdots \\ \dot{z}_{k-1} &= -\tilde{\lambda}_1 L^{\frac{k}{k+1}} [z_0 - f(t)]^{\frac{1}{k+1}} + z_k, \\ \dot{z}_k &= -\tilde{\lambda}_0 L \operatorname{sign}(z_0 - f(t)). \end{aligned} \quad (3.8)$$

with the notation

$$[w]^\gamma = |w|^\gamma \operatorname{sign}(w), \quad (\text{and } [w]^0 = \operatorname{sign}(w)).$$

They also used Equivalent Control theorem to extract the equivalent control $u_{\text{eq}}(t)$ from a discontinuous SM control signal $u(t)$. They developed a nonlinear filtering approach to filter discontinuities, and yield estimates of $u_{\text{eq}}(t)$ and its derivatives [11].

$$\begin{aligned} s^{(r)} &= h(t, x) + g(t, x) u \\ &= g(t, x) \left(u - u_{\text{eq}}(t, x) \right), \end{aligned} \quad \text{with} \quad u_{\text{eq}}(t, x) = -\frac{h(t, x)}{g(t, x)}. \quad (3.9)$$

where $s(t, x)$ is output and u is the control input are scalars and the system has a relative degree r .

$$\begin{aligned} \dot{z}_{-2} &= u(t) - \gamma z_{-2}, \\ \dot{z}_{-} &= \mathbf{D}_k(z_{-}, z_{-2}, L, \lambda) - e_0 \gamma z_{-2} \\ \mathbf{D}_k(z_{-}, z_{-2}, L, \lambda) &= dz_{i+1} = -\lambda_{k+1-i} L^{\frac{i+1}{k+1}} \left(|\text{error}|^{\frac{k-i}{k+1}} \text{sign}(\text{error}) \right) + z_{i+2} \end{aligned} \quad (3.10)$$

They also provided the discrete version of the method for practical application [using eqs. 22, 23, 26, & 27 in the paper]. Here are eqs. 22 and 23, which are supposed to provide the same result. The former is based on discrete extraction of equivalent control, and the later is based on the proposed novel filter in discrete form.

For a sampling instants t_j , the differentiator is discretized as

$$z(t_{j+1}) = z(t_j) + D_k(z(t_j), f(t_j), L, \lambda) \tau_j + T_k(z(t_j), \tau_j), \quad (3.11)$$

where $T_k(z(t_j), \tau_j)$ is a Taylor-like correction term (its explicit form is provided in the paper) that ensures the discrete dynamics maintain the homogeneous error properties.

Similarly, the discrete version of the equivalent control extraction filter is given by

$$\begin{aligned} z_{-2}(t_{j+1}) &= e^{-\gamma \tau_j} z_{-2}(t_j) + \frac{1}{\gamma} (1 - e^{-\gamma \tau_j}) u(t_j), \\ z_{-}(t_{j+1}) &= z_{-}(t_j) + \left[D_k(z_{-}(t_j), z_{-2}(t_j), L, \lambda) - e_0 \gamma z_{-2}(t_j) \right] \tau_j \\ &\quad + T_k(z_{-}(t_j), \tau_j). \end{aligned} \quad (3.12)$$

3.3 Numerical differentiation with annihilators

Mboup, Join, and Fliess (2009), who proposed an operational calculus-based differentiation scheme and their technique was based on differential algebra and utilized iterated integrals to

extract derivative estimates [2].

They considered the estimation of $x^{(n)}(t)$, the n^{th} order derivative of a smooth signal $x(t)$ defined on an interval $\mathcal{I} \subset \mathbb{R}_+$. The signal $x(t)$ is not directly available but rather it is observed through a noise measurement. The $\varpi(t)$ denote the noise and the $y(t) = x(t) + \varpi(t)$ is the expression of the noisy observation. The problem of differentiation is challenging since the direct differentiation tends to amplify the noise.

$x(t)$ is analytic on $\mathcal{I}$, the convergent Taylor expansion at $t = 0$

$$x(t) = \sum_{i \geq 0} x^{(i)}(0) \frac{t^i}{i!}. \quad (3.13)$$

In real applications, using an infinite series is not feasible, it is essential to approximate with a finite number of terms. With $N \geq n$, the truncated Taylor expansion is

$$x_N(t) = \sum_{i=0}^N x^{(i)}(0) \frac{t^i}{i!}, \quad (3.14)$$

In the frequency domain, it becomes

$$s^{N+1} \hat{x}_N(s) = s^N x(0) + s^{N-1} \dot{x}(0) + \dots + s^{N-n} x^{(n)}(0) + \dots + x^{(N)}(0). \quad (3.15)$$

Taylor series expansion provided a local approximation of the signal. This expansion was then transformed into the operational domain using Laplace transforms, where differentiation became an algebraic problem. The operational domain representation transforms the problem into a solvable algebraic form.

3.3.1 Annihilator equation

Estimating all Taylor coefficients $x^{(i)}(0)$ simultaneously can introduce noise and numerical errors. The terms $s^{N-i} x^{(i)}(0)$ in the above equation, for $i \neq n$ act as perturbations. The purpose of annihilator is to eliminate these terms and focus only on $x^{(n)}(0)$. These annihilators removed higher-order terms that could contribute to estimation errors.

The linear differential operator Π , defined as:

$$\Pi \hat{x}_N = \varrho(s) x^{(n)}(0), \quad (3.16)$$

where $\varrho(s)$ is some rational function in $\mathbb{C}(s)$.

$$f(t)\tilde{x}_N^{(n)}(0) = \sum_{\text{finite}} \int_0^t h_\ell(t-\alpha)(-1)^\ell \alpha^\ell y(\alpha) d\alpha. \quad (3.17)$$

The estimated derivative is expressed as an integral operator in noisy observations. The integral formulation expressed the derivative as a weighted sum of the observed noisy signal over a short time interval.

Pointwise Derivative Estimation Using Orthogonal Polynomials

$$\tilde{x}^{(n)}(0; \kappa, \mu) = \gamma_{\kappa, \mu, n} \int_0^1 \tau^{\kappa+n} (1-\tau)^{\mu+n} x^{(n)}(T\tau) d\tau \quad (3.18)$$

$$+ (-1)^n \frac{\gamma_{\kappa, \mu, n}}{T^n} \int_0^1 \frac{d^n}{d\tau^n} \{ \tau^{\kappa+n} (1-\tau)^{\mu+n} \} \varpi(T\tau) d\tau.$$

To construct robust estimators of numerical differentiation using Jacobi orthogonal polynomials. This equation provides estimates of the true derivative while accounting for noise effects.

3.3.2 Time-Delayed Derivative Estimation

$x(t)$ be observed through a noise corruption $\varpi(t)$ by $y(t) = x(t) + \varpi(t)$. The causal n^{th} order minimal (κ, μ) -algebraic numerical differentiation of $y(t)$ is time-delayed:

$$\tilde{x}^{(n)}(t; \kappa, \mu) \approx x^{(n)}(t - \tau_1), \quad t \geq T. \quad (3.19)$$

The delay τ_1 given as $\tau_1 = T\xi_1$ where ξ_1 is given below.

$$\xi_1 = \frac{\kappa + n + 1}{\mu + \kappa + 2(n + 1)} \quad (3.20)$$

A digital FIR filter was also developed by the authors and given in equation: [eqs. 48, 49 in the paper]

$$\tilde{x}^{(n)}(t_\ell; \kappa, \mu; N) \approx \tilde{x}_\ell^{(n)} = \sum_{m=0}^M W_m g_m y_{\ell-m}. \quad (3.21)$$

This equation was supposed to implement numerical differentiation in discrete time using Finite Impulse Response (FIR) digital filters.

3.4 Our contribution

It is well established that the High-Gain Observer (HGO) and High-Gain Controller (HGC) methods tend to amplify noise when the noise level is sufficiently strong. The differentiation methods proposed by Ibrir [1] also exhibited this issue. After a thorough analysis of his approach, we proposed a modified method that enhances the accuracy of noisy signal differentiation. Specifically, our proposed modification simultaneously mitigates noise amplification and reduces the peaking phenomenon. The following section details the improvements introduced in our modified approach.

$$\begin{aligned}\dot{x}_i(t) &= x_{i+1}(t) + C_n^i (t/a + e)^i \gamma^i (y(t) - x_1(t)), \quad i = 1, \dots, n-1, \\ \dot{x}_n(t) &= C_n^n (t/a + e)^n \gamma^n (y(t) - x_1(t)).\end{aligned}\tag{3.22}$$

where C_n^i are the binomial coefficients

t - time bound by $\tanh()$ and

a - is a correction factor and its value should be greater than zero, which is a compromise between peak and accuracy

e - epsilon to get non-zero gain to $t = 0$

We have also proposed a modification for the HGC method that gave us a better result. The original method proposed by [1] has two versions, and both versions have a problem. The first one didn't eliminate the peak phenomenon at the beginning, which is very large, even though it's better than HGO. The second version eliminated the peak phenomenon; however, the result grows unboundedly to infinity as time goes on. The proposed method solves both problems and also improves the noise reduction by far a lot. The equation after modification is given as follows:

$$\begin{aligned}\dot{x}_i(t) &= x_{i+1}(t), \quad 1 \leq i \leq n-1, \\ \dot{x}_n(t) &= -(\gamma(t/a) + e)^n (x_1(t) - y(t)) - \sum_{i=1}^{n-1} C_n^i (\gamma(t/a) + e)^i x_{n-i+1}(t),\end{aligned}\tag{3.23}$$

Why our proposed method is better?

The proposed method solves most of the problems that exist in the original work. For small noises, the result is comparable to the sliding mode differentiation technique, which is supposed to be more robust and noise insensitive. Additionally, the introduction of a correction factor

provides flexibility in tuning the gains for different signal characteristics without requiring any parameter modifications.

Why we did all those complex calculations?

3.5 Model-free control with noisy signal differentiation

As discussed in the introduction and literature review sections, noisy signal differentiation is essential for estimating derivatives from noisy measurements. These differentiations are necessary for controlling plants with uncertain dynamics.

In this case, we prefer to design a controller without requiring detailed knowledge of the plant's internal structure, relying solely on real-time available measurements, similar to a black-box approach. However, in practical applications, this black-box approach can pose risks, particularly in critical systems such as aviation or surgical robotics. Therefore, our approach must incorporate known system dynamics while simultaneously estimating uncertainties to ensure stability and reliability.

Model-free control offers advantages in systems with unknown or complex dynamics, as it eliminates the need for precise mathematical models [9], [10]. High-gain observers effectively estimate system states and derivatives, even in the presence of noise [11], [12], while sliding mode control enhances robustness against uncertainties and disturbances [11], [10].

High-gain observers (HGOs) play a crucial role in model-free control by providing accurate estimations of system states and their derivatives, which are often unavailable or difficult to measure directly [11]. This is particularly valuable in situations where precise system models are unavailable or computationally expensive to obtain [10]. The HGO's ability to handle noise is a key advantage, enabling its application in real-world scenarios where measurement imperfections are common [12]. Zhenshuai Wan et al. [11] demonstrated the effectiveness of an HGO in conjunction with a sliding mode controller (SMC) for an electro-hydraulic servo system, achieving improved control performance despite parameter uncertainties and external disturbances. The accurate estimation of the position signal's time derivative, provided by the HGO, was essential for the SMC design.

In high gain observer method, the estimation of states can be done using the concept of extended state and its effectiveness strongly depends on a quality of estimate of the extended state $\hat{f}$.

This extended state is the sum of any uncertainties, disturbance and noise. The total disturbance can be estimated by treating f as an additional state variable of the plant. Hence, new variable $f(\cdot)$ together with a real state of a plant constitute the so-called extended state (ES)[13].

The ES for a plant dynamics can be written as $\ddot{y} = f(\cdot) + \hat{b}u$ for second order system [13].

$$x_1 \triangleq y, \quad (3.24)$$

$$x_2 \triangleq \dot{y}, \quad (3.25)$$

$$x_3 \triangleq f(\cdot) \quad \leftarrow \text{additional state variable} \quad (3.26)$$

The approach of this method is active disturbance rejection control (ADRC) method which is based on Luenberger Observer method as discussed in lecture materials of Poznan University of Technology [13].

However, the complete model-free approach approximate any model into a first order model as:

$$y \triangleq \text{measured output} \quad (3.27)$$

$$\dot{y} \triangleq f(\cdot) + \alpha u \quad (3.28)$$

where u is the control input and f is the unknown dynamics

The state to be estimated is f , which is unknown a priori. Using the ultra-local model as proposed by Yahagi et al [14], we can estimate this unknown dynamics as follows:

$$\begin{aligned} e &= y_r - y \\ \dot{e} + ke &= 0 \\ \dot{y}_r - \dot{y} + ke &= 0 \\ \dot{y}_r - \hat{f}(\cdot) - \alpha u + ke &= 0 \\ u &= \dot{y}_r - \hat{f}(\cdot) + ke \rightarrow P \text{ control only} \\ u &= \dot{y}_r - \hat{f}(\cdot) + ke + k_i \int e dt \rightarrow PI \\ u &= \ddot{y}_r - \hat{f}(\cdot) + k_p e + k_d \dot{e} \rightarrow PD \\ u &= \ddot{y}_r - \hat{f}(\cdot) + k_p e + k_i \int e dt + k_d \dot{e} \rightarrow PID \end{aligned} \quad (3.29)$$

The error e is given as a difference of the reference input and the estimated state. The importance

of our noisy signal differentiation technique is to get the best possible estimate of the signal in noisy environments. The estimated states will be found using the HGO estimation method. The block diagram of this method is given as follows.

This proposed MFC model was given by Yahagi et al [14]

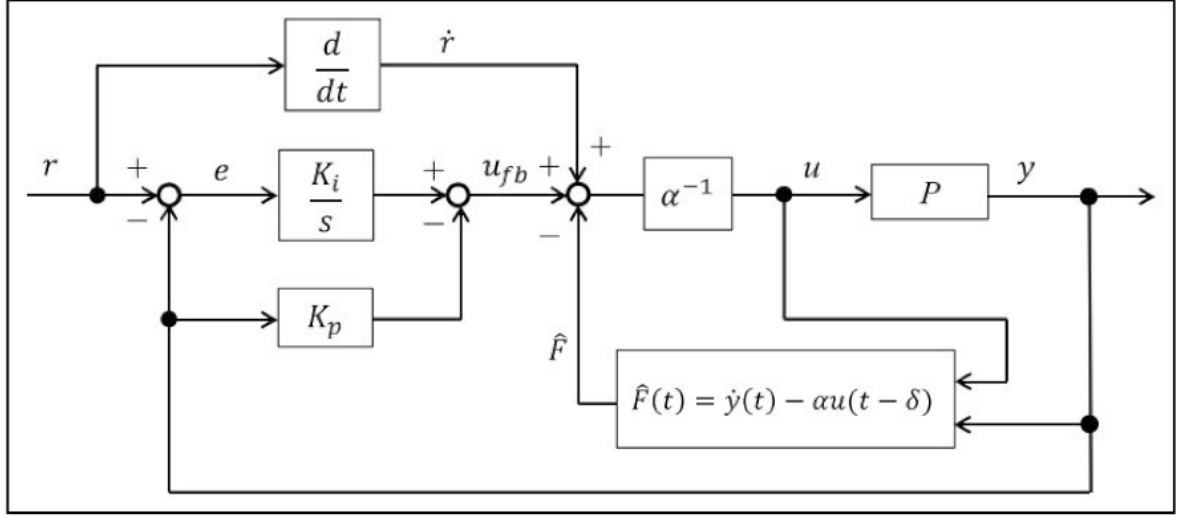


Figure 3.1: Model-free control (MFC) with an ultra-local model

This model is also known as intelligent PI model.

CHAPTER 4: SIMULATIONS, RESULTS AND DISCUSSIONS

For the simulation of the signal differentiation techniques studied, we chose two signals and two noises with different parameters and compared the results. The selected signals that are expected to be available in real-time from the plant are given by

$$y_1(t) = \frac{-\cos(t)}{2 + \sin(3t)}$$

and

$$y_2(t) = \frac{-(1 + \exp(-2t)) \cdot \sin(2t)}{5 + \cos(t)}$$

and the white noise with variance of $1e-8$ and uniform random generator of magnitude $[-1e-3, 1e-3]$. In both cases, we have seen an improvement for the noise amplification, and we added a discussion under each figure. Finally, we added simulation results of our contribution that improve the S. Ibrir's HGO & HGC methods. In our work, noise is reduced even as the order of differentiation increases.

4.1 High-Gain Observers for Signal Differentiation

4.1.1 Brunovski form: High Gain Controllable (HGC) canonical form

To understand the concept of Brunvski form, we used equation 3.5 and implemented it in Simulink for two different signals and two different noises. In both cases, the method reduced the noise amplification compared to the classical differentiation of noisy signals.

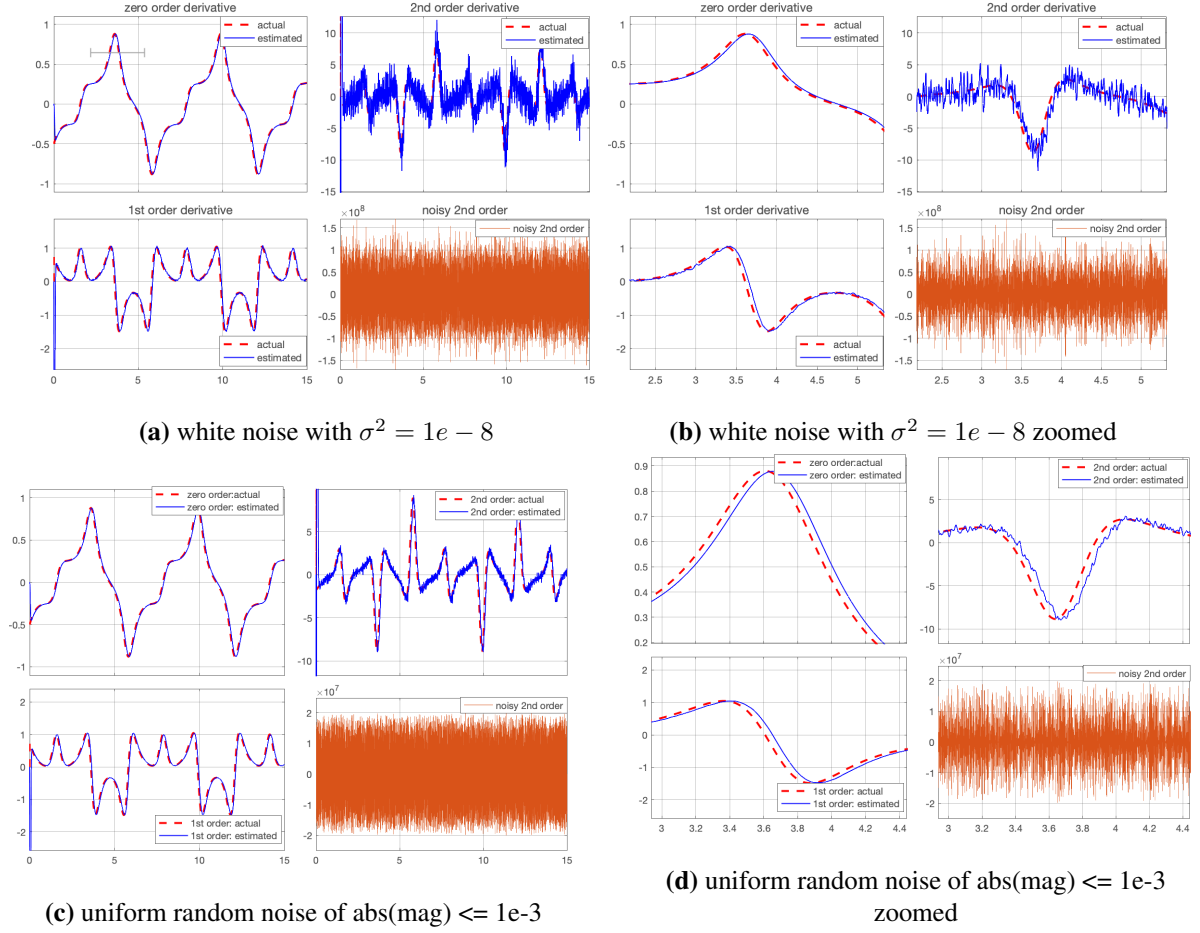
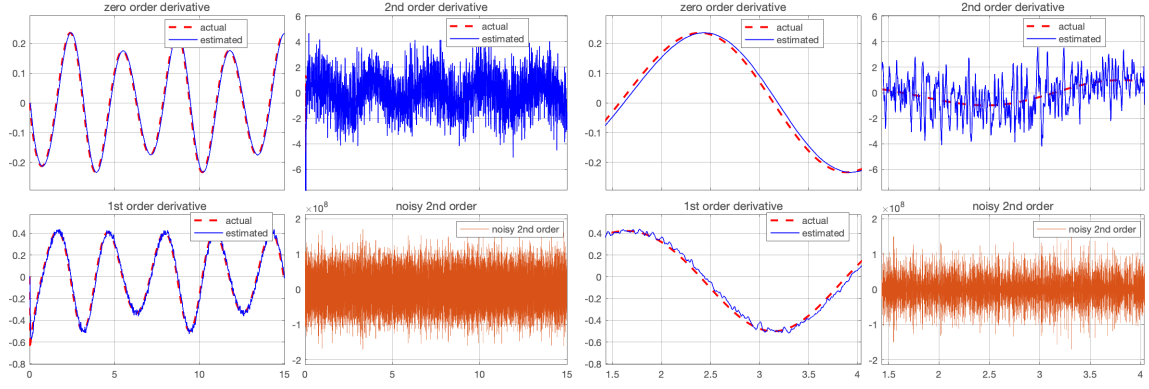


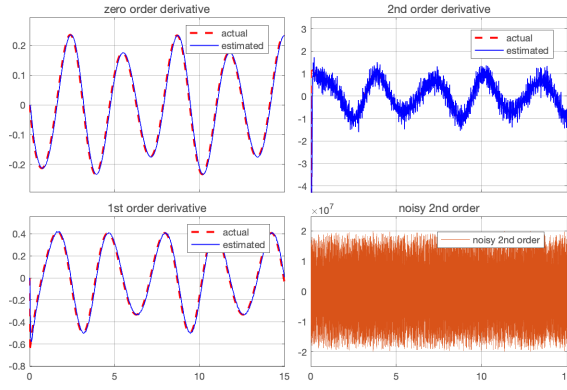
Figure 4.1: Brunovski form estimation for $y_1(t)$

As shown in Figure 4.1, we used the Brunovski form of the high gain to estimate the function $y_1(t) = -\cos(t)/(2 + \sin(3t))$ and its derivative in the presence of noise. Compared with the observable canonical form, this method reduced the peaking phenomena and provided good estimation but less accurate than the former. As the order of differentiation increases, the noise sensitivity also increases. However, appropriate parameter tuning (via γ) helps mitigate this issue by balancing trade-off between accuracy, noise sensitivity, and peaking behaviour. The results demonstrated the ability of the Brunovski form to reduce peaking better than observable high-gain differentiation. For this simulation, we have taken $\varepsilon = \frac{1}{90}$ i.e. $\gamma = 90$ and $k_i = C_3^i$.

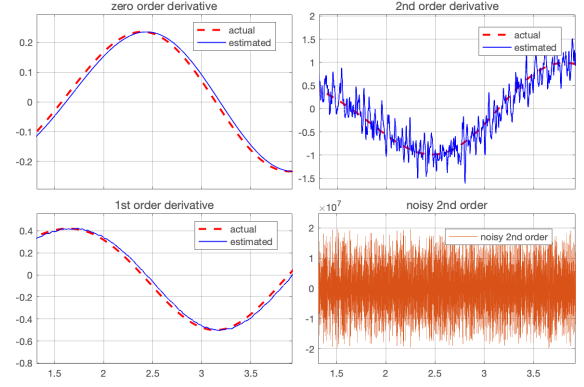


(a) white noise with $\sigma^2 = 1e - 8$

(b) white noise with $\sigma^2 = 1e - 8$ zoomed



(c) uniform random noise of $\text{abs}(\text{mag}) \leq 1e-3$



(d) uniform random noise of $\text{abs}(\text{mag}) \leq 1e-3$ zoomed

Figure 4.2: Brunovski form estimation for $y_2(t)$

Similarly, as shown in fig 4.2, we tested another function $y_2(t) = -(1 + e^{-2t}) \sin(2t)/(5 + \cos(t))$ to further evaluate the performance of the observer yielding results consistent with our previous observations. The Brunovski form effectively reduced the peaking phenomenon while maintaining good accuracy, even as noise sensitivity increased with differentiation order. The primary issue in this form is the initial error, which is higher than that observed in the High-Gain Observer (HGO), despite decreasing over time.

4.1.2 High Gain Observable (HGO) canonical form

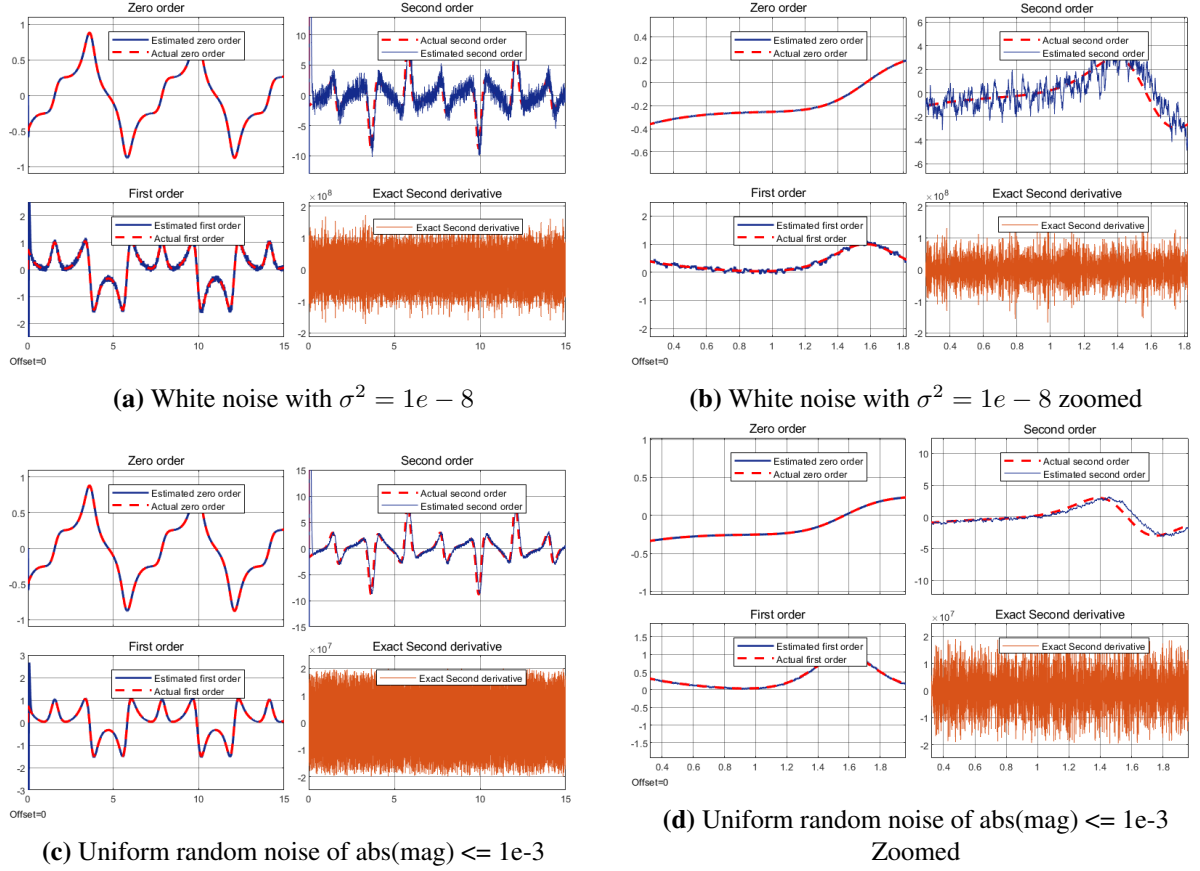
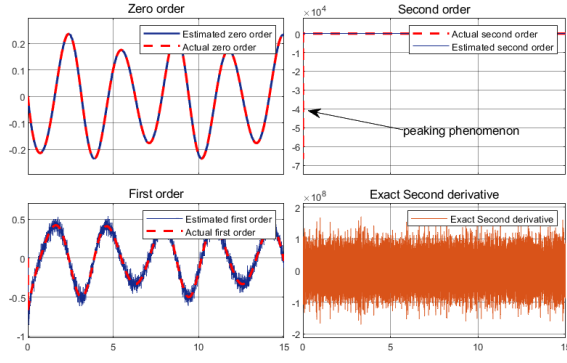
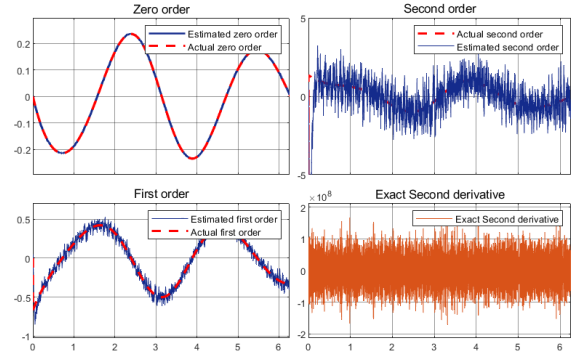


Figure 4.3: High-gain observer estimation for $y_1(t)$

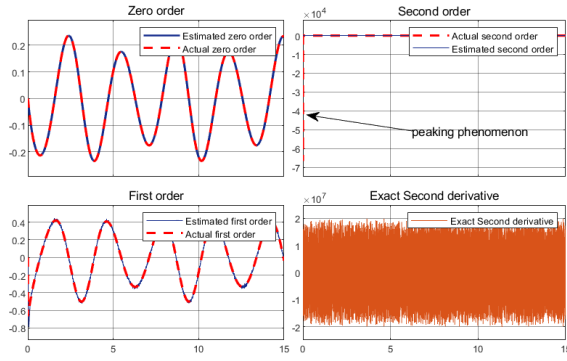
As shown in Figure 4.3, we used the HGO form of the high gain to estimate the function $y_1(t) = -\cos(t)/(2 + \sin(3t))$ and its derivative in the presence of noise. Compared with the Brunovski form, this method has a peaking phenomenon and while providing accurate estimation than Brunovski method. As the order of differentiation increases, the noise sensitivity also increases. However, appropriate parameter tuning (via γ) helps mitigate this issue by balancing trade-off between accuracy, noise sensitivity, and peaking behaviour. The results demonstrated the ability of the HGO form to provide accurate results, especially in noise free environments. For this simulation, we have taken $\varepsilon = \frac{1}{50}$ i.e. $\gamma = 50$ and $k_i = C_3^i$.



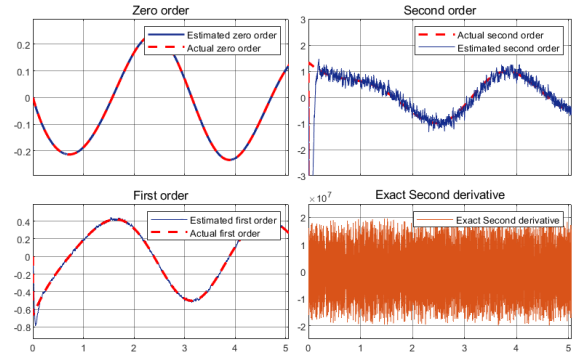
(a) White noise with $\sigma^2 = 1e - 8$



(b) White noise with $\sigma^2 = 1e - 8$ zoomed



(c) Uniform random noise of $\text{abs}(\text{mag}) \leq 1e-3$

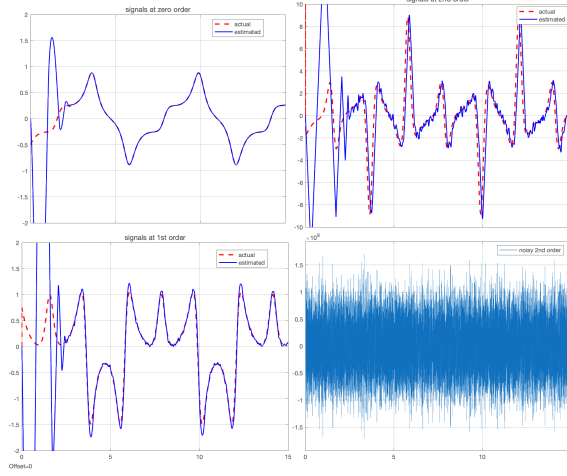


(d) Uniform random noise of $\text{abs}(\text{mag}) \leq 1e-3$ Zoomed

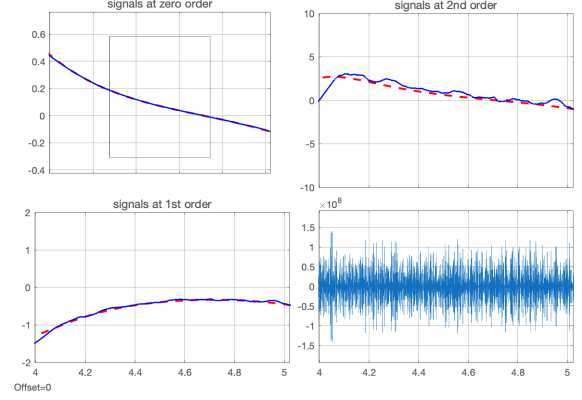
Figure 4.4: High-gain observer estimation for $y_2(t)$

Similarly, as shown in fig 4.4, we tested another function $y_2(t) = -(1 + e^{-2t}) \sin(2t)/(5 + \cos(t))$ to further evaluate the performance of the HGO method yielding results consistent with our previous observations. The HGO form gave us good accuracy for the new signal even as noise sensitivity increased with differentiation order. So, we can conclude that for different signals, HGO method performs with a good accuracy even though the peaking phenomenon is a critical problem.

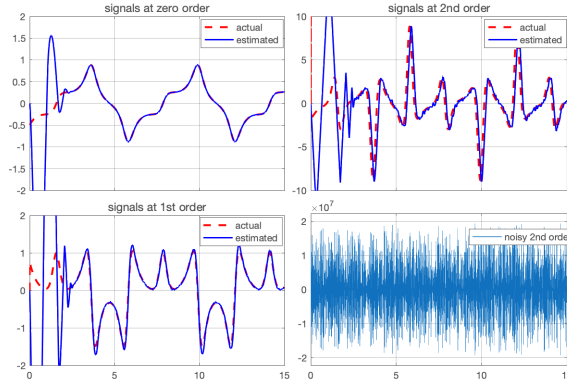
4.2 Sliding-Mode-Based Differentiation and its Application



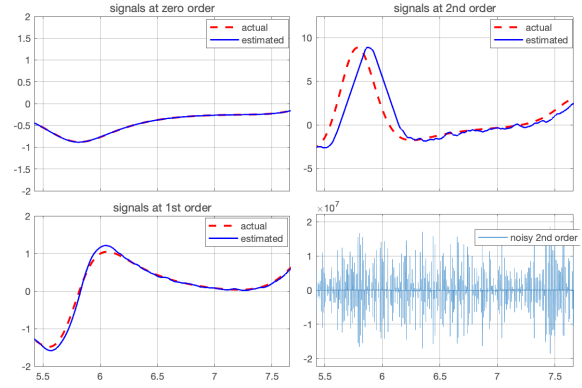
(a) White noise with $\sigma^2 = 1e - 8$



(b) White noise with $\sigma^2 = 1e - 8$ zoomed



(c) Uniform random noise of $\text{abs}(\text{mag}) \leq 1e-3$

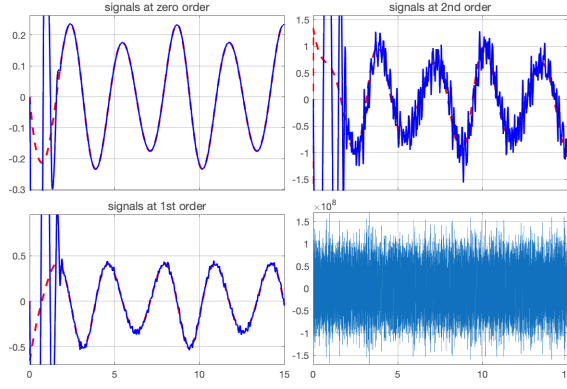


(d) Uniform random noise of $\text{abs}(\text{mag}) \leq 1e-3$
Zoomed

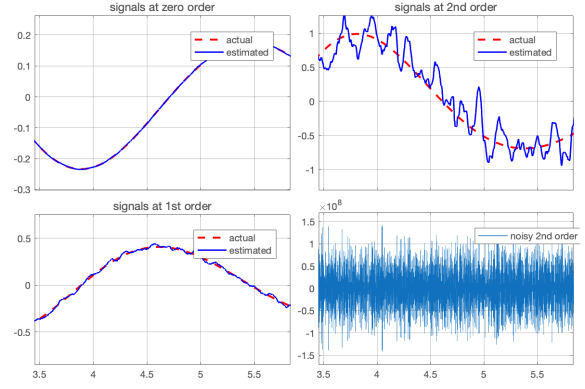
Figure 4.5: HOSM estimation for $ds = y_1(t)$

Figures 4.5 and 4.6 (below) demonstrate the actual and the estimated derivatives obtained using HOSM. Compared to the results obtained from HGO, this novel Sliding Mode Differentiator shows improved performance with no peaking phenomenon, ensuring that the estimated derivative converges to the actual derivative within finite time.

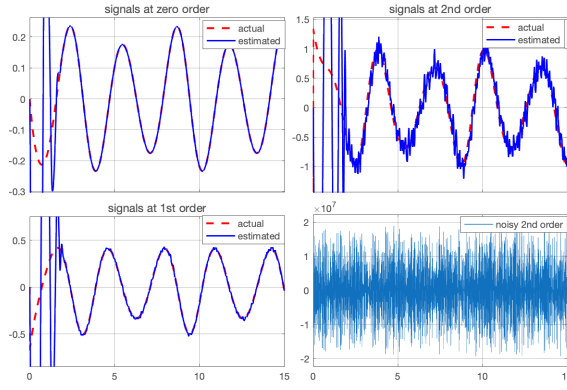
Additionally, compared to classical sliding mode differentiation, the HOSM has higher-order dynamics and continuous approximation; this effectively reduces the chattering effect, leading to a more stable and accurate differentiation process, even in the presence of different noises. We tested this method for the signal $y_1(t) = -\cos(t)/2 + \sin(3t)$ and the result in Figure 4.5 was found by adding white noise with variance 1×10^{-8} and uniform random noise in the range $[-1e - 3, 1e - 3]$.



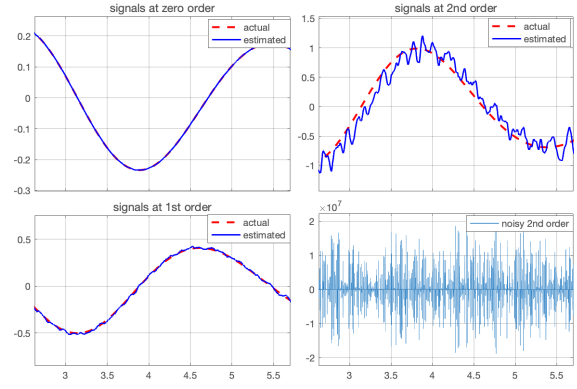
(a) White noise with $\sigma^2 = 1e - 8$



(b) White noise with $\sigma^2 = 1e - 8$ zoomed



(c) Uniform random noise of $\text{abs}(\text{mag}) \leq 1e-3$



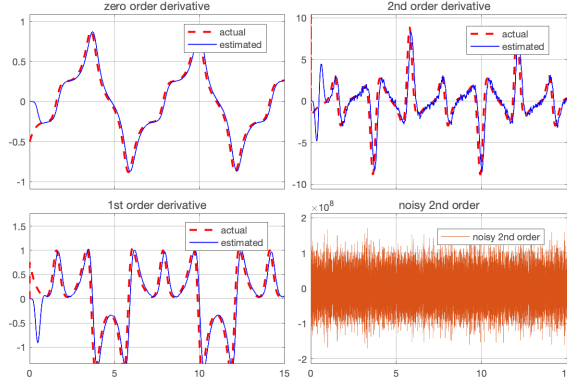
(d) Uniform random noise of $\text{abs}(\text{mag}) \leq 1e-3$
Zoomed

Figure 4.6: HOSM estimation for $ds = y_2(t)$

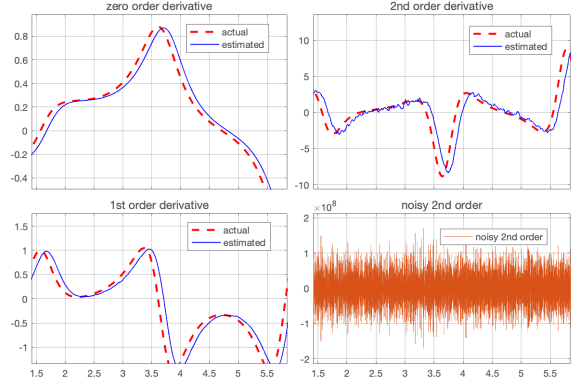
We have also tested this method for the signal, $y_2(t) = -(1 + e^{-2t}) \sin(2t)/(5 + \cos(t))$ and obtained the results presented in Figure 4.6 by adding white noise with a variance of 1×10^{-8} and uniform random noise in the range $[-10^{-3}, 10^3]$. This result is one of the promising outcomes necessary for feedback control systems. Furthermore, the results in Figures 4.5 and 4.6 demonstrate that the Higher-Order Sliding Mode (HOSM) differentiator exhibits greater robustness in the presence of measurement noise, making it less sensitive to disturbances.

4.3 Our Contribution

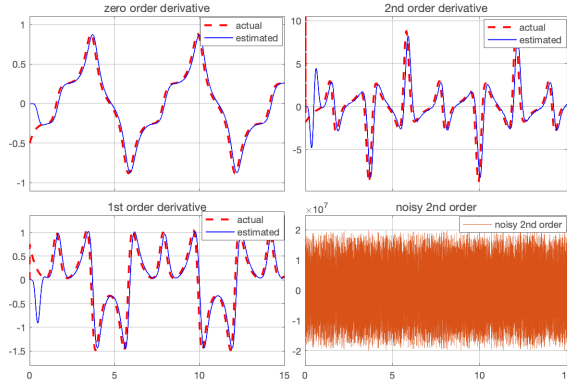
4.3.1 Modified Brunovski Form (MHGC) canonical form



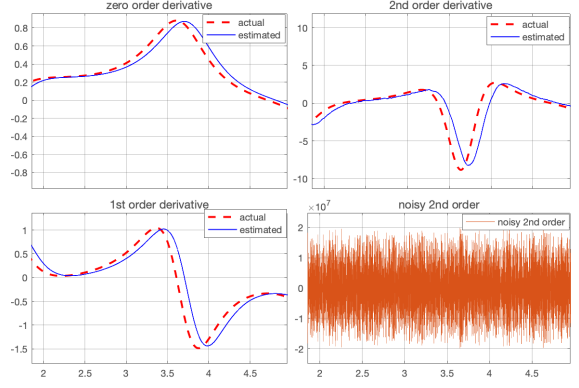
(a) White noise with $\sigma^2 = 1e - 8$



(b) White noise with $\sigma^2 = 1e - 8$ zoomed



(c) Uniform random noise of $\text{abs}(\text{mag}) \leq 1e-3$

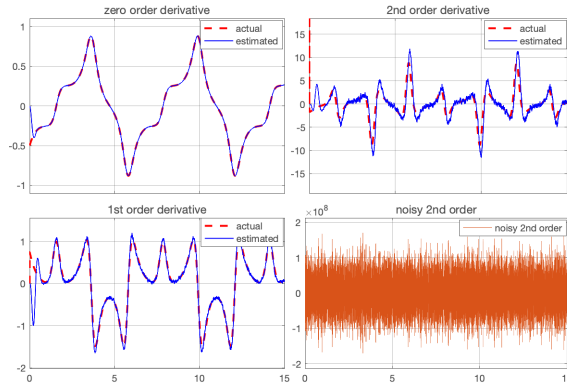


(d) Uniform random noise of $\text{abs}(\text{mag}) \leq 1e-3$ zoomed

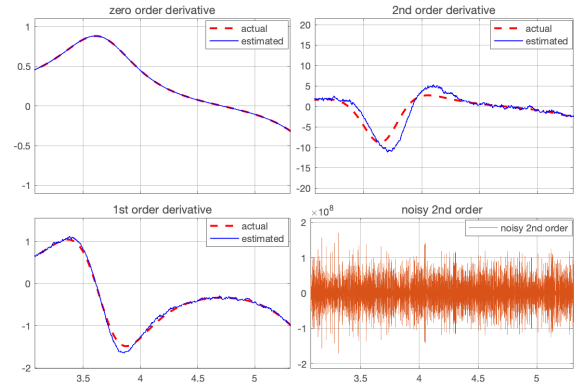
Figure 4.7: Modified Brunovski form high gain estimation for $y_1(t)$

The modified Brunovski form effectively addresses both the peaking phenomenon observed in the original formulation, although with less amplification of noise than in the High-Gain Observer (HGO), and the issue of unbounded response growth due to noise amplification. This method not only eliminates the peaking phenomenon entirely but also significantly reduces both noise sensitivity and amplification. As demonstrated in Figure 4.8, the proposed approach yields superior results compared to the method introduced by S. Ibrir [1].

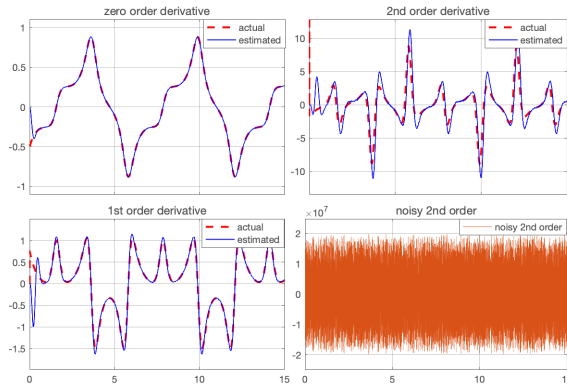
4.3.2 Modified High Gain Observable (MHGO) canonical form



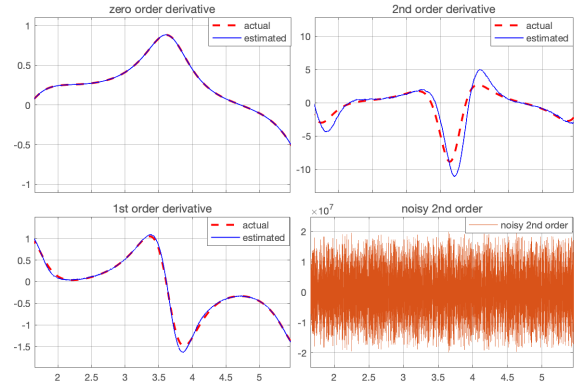
(a) White noise with $\sigma^2 = 1e - 8$



(b) White noise with $\sigma^2 = 1e - 8$ zoomed



(c) Uniform random noise of $\text{abs}(\text{mag}) \leq 1e-3$



(d) Uniform random noise of $\text{abs}(\text{mag}) \leq 1e-3$
Zoomed

Figure 4.8: Modified High-gain observer estimation for $y_1(t)$

The modified High-Gain Observer (HGO) method was tested on the signal $y_1(t) = -\cos(t)/(2 + \sin(3t))$ with two different noise conditions and effectively addressed both the peaking phenomenon observed in the original formulation and the issue of high noise amplification. This method successfully eliminated the peaking phenomenon while significantly reducing noise sensitivity and amplification. This method eliminated the peaking phenomenon as well as reduced the noise sensitivity and amplification. As shown in Fig. 4.9, the results are better than the one proposed by S. Ibrir [1]. The error is also very small even in the presence of noise. The result showed as it is comparable with HOSM. These results showed the effectiveness of our modified proposition.

4.4 Model free control with noisy signal differentiation

To evaluate the effectiveness of the signal differentiation technique in estimating the unknown dynamics of a system, we selected the roll dynamics of an air-plane, incorporating both known and unknown components. The objective was to track the known dynamics while simultaneously estimating the unknown dynamics. In this study, we aim to track the first derivative as well, employing a grey-box approach. This method, referred to as model-free control (MFC) with a second-order approximation, offers improved performance compared to a fully model-free control strategy.

Dynamics of the plant: We have chosen a plane dynamics and approximated the roll-angle dynamics by the following nonlinear differential equation.

$$\ddot{\beta} = \theta_{10}\beta + \theta_{20}\dot{\beta} + \theta_{60}(\delta_a + d) + (\theta_{30}|\beta| + \theta_{40}|\dot{\beta}|)\dot{\beta} + \theta_{50}\beta^3, \quad (4.1)$$

where $\theta_{10}, \theta_{20}, \dots, \theta_{60}$ are the true parameters of the plant, while the term d represents some external disturbance which affects the plant through the input channel, this indicates that the system is highly uncertain, and the addition of noise can lead to severe instability and performance degradation. The only measurable signals are the control input $u = \delta_a$ and the plant output, that is, the roll angle: $y = \beta$.

The equations are rewritten by decomposing equation 4.1 into known and unknown components.

$$\begin{aligned} \ddot{\beta} &= \theta_{10}\beta + \theta_{20}\dot{\beta} + \hat{\theta}_6\delta_a + F(\beta, \dot{\beta}, \delta_a, d) \\ F(\beta, \dot{\beta}, \delta_a, d) &= (\theta_{30}|\beta| + \theta_{40}|\dot{\beta}|)\dot{\beta} + \theta_{50}\beta^3 + \theta_{60}d + (\theta_{60} - \hat{\theta}_6)\delta_a \end{aligned} \quad (4.2)$$

The unknown dynamics estimated $F(\beta, \dot{\beta}, \delta_a, d)$ using one of our proposed methods. To evaluate the feasibility of estimating the unknown dynamics, we applied our modified estimation method. The results are presented below.

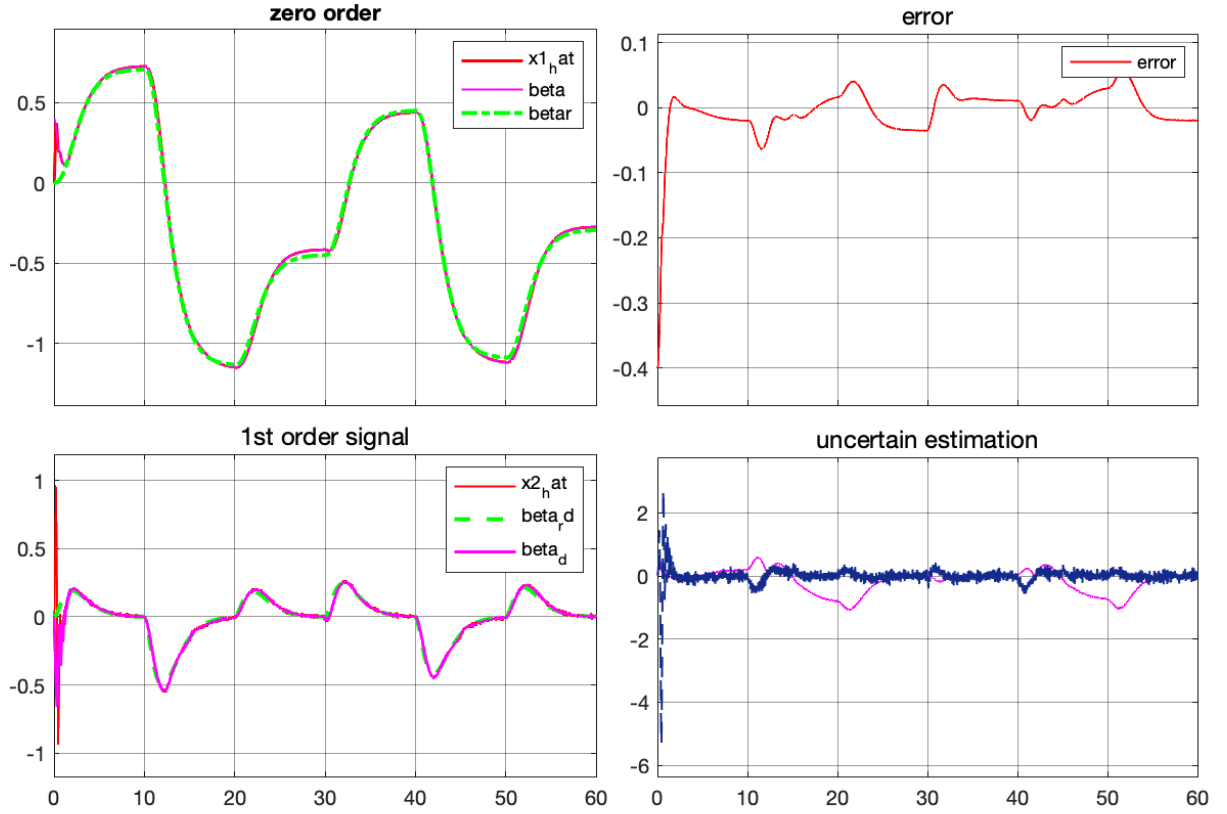


Figure 4.9: Model-free control with modified HGO

As shown in Fig. 4.9, the controller tracked the known dynamics while estimating the unknown dynamics. Even though there are some gaps between the actual uncertainty and the estimated one, it is a good approximation, and we are working on improving the estimation.

From this, we can conclude that the model-free control approach, when integrated with signal differentiators, provides an effective control strategy for systems with uncertain or partially known dynamics. By incorporating advanced techniques for handling noisy signal measurements, such as high-order sliding mode differentiators and high-gain observers, the proposed approach enhances state estimation accuracy while minimizing the impact of noise. This combination improves the robustness and reliability of control performance, making it a viable solution for applications where precise model information is unavailable or difficult to obtain.

CHAPTER 5: CONCLUSION AND RECOMMENDATION

5.1 Conclusion

In this project, we investigated advanced differentiation methods designed to estimate derivatives from noisy measurements, a main challenge in control applications and signal processing. High-gain observer methods, higher-order sliding mode differentiators, and numerical differentiation with annihilators were reviewed with detailed analysis. We implemented the two approaches: High-Gain Observers (HGO) and High-Order Sliding Mode Differentiators (HOSMDs). Each method was analyzed in terms of accuracy, noise robustness, and computational efficiency. Through MATLAB/Simulink simulations, we demonstrated that while traditional high-gain observers offer fast convergence, they suffer from noise amplification and the peaking phenomenon. In contrast, sliding mode differentiators, particularly high-order formulations, exhibited robustness against noise while ensuring finite-time convergence.

In addition to applying existing methods, our contribution lies in modifying the original proposition of high gain observer differentiation method by introducing hyperbolic tangent and correction factor to choose a compromise between peaking and accuracy. This modification improved the overall robustness of the signal differentiation, leading to more accurate results in noisy environments which is almost comparable with higher-order sliding mode differentiator. For different noises and signals, we can adjust the performance using the correction factor.

Finally, we used the modified version of high gain observer to estimate the uncertainty in model-free control design. The differentiator estimated the uncertainty, which led to favourable outcomes, further validating the approaches, however it needs an improvement for more accurate estimation.

5.2 Recommendation

The next step in this research will be to test the signal differentiation techniques in the IBSC lab, allowing for a more comprehensive evaluation of their effectiveness in real-world applications. This will help determine their viability in practical scenarios and highlight areas for potential improvement.

Additionally, we plan to review and explore other advanced techniques in signal differentiation, comparing their performance against the methods used in this study to see if there are opportunities for further enhancement.

Finally, a key direction for future work will be to focus on improving the accuracy of signal estimation within the context of model-free control. This will contribute to enhancing the overall performance of the system when differentiating noisy signals in complex environments.

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APPENDIX: SIMULINK MODELS

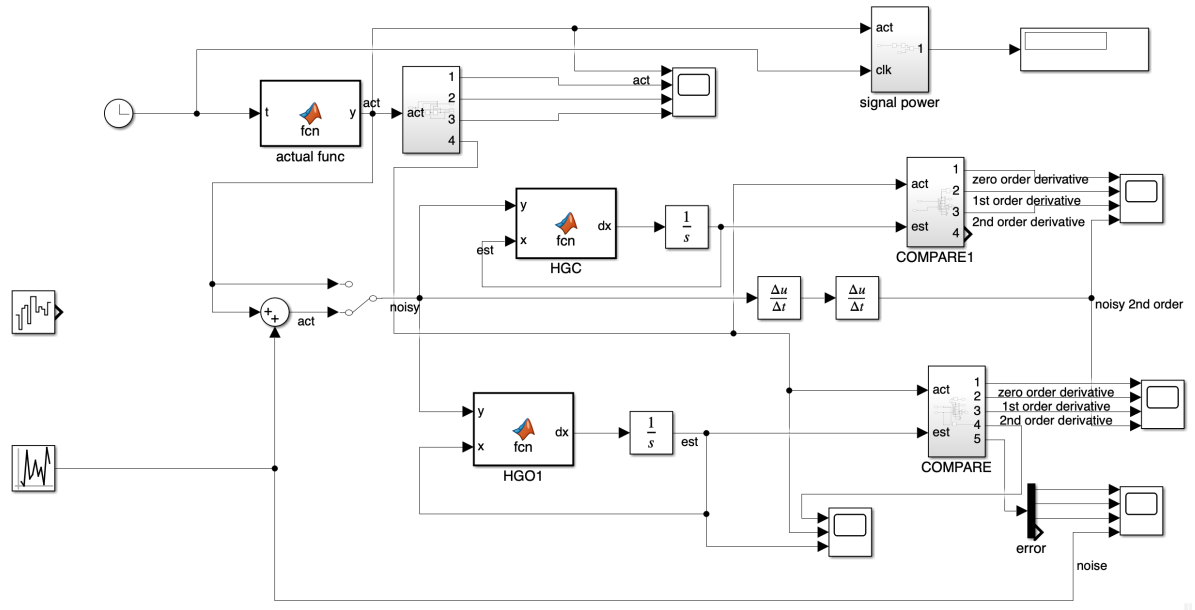


Figure 5.1: HGO & HGC Simulink models

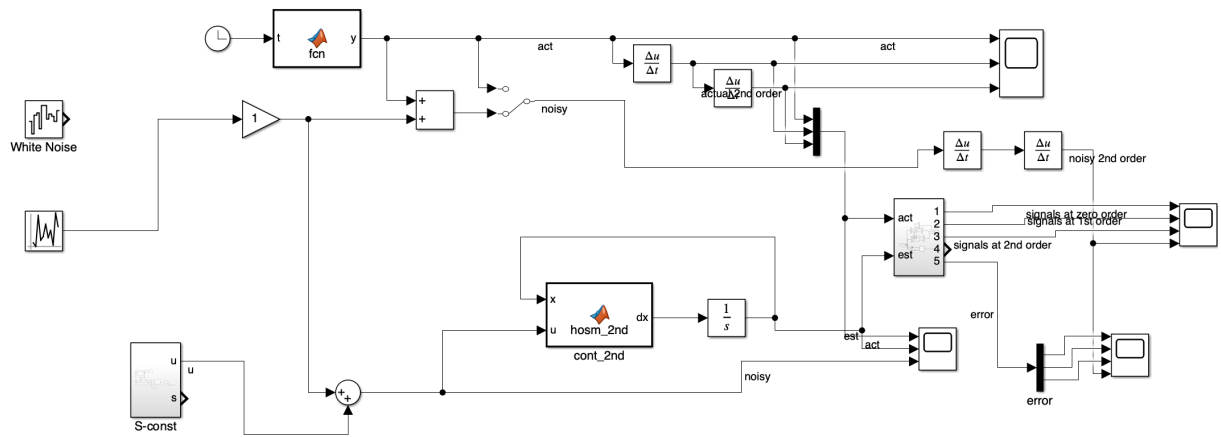


Figure 5.2: HOSM Simulink models

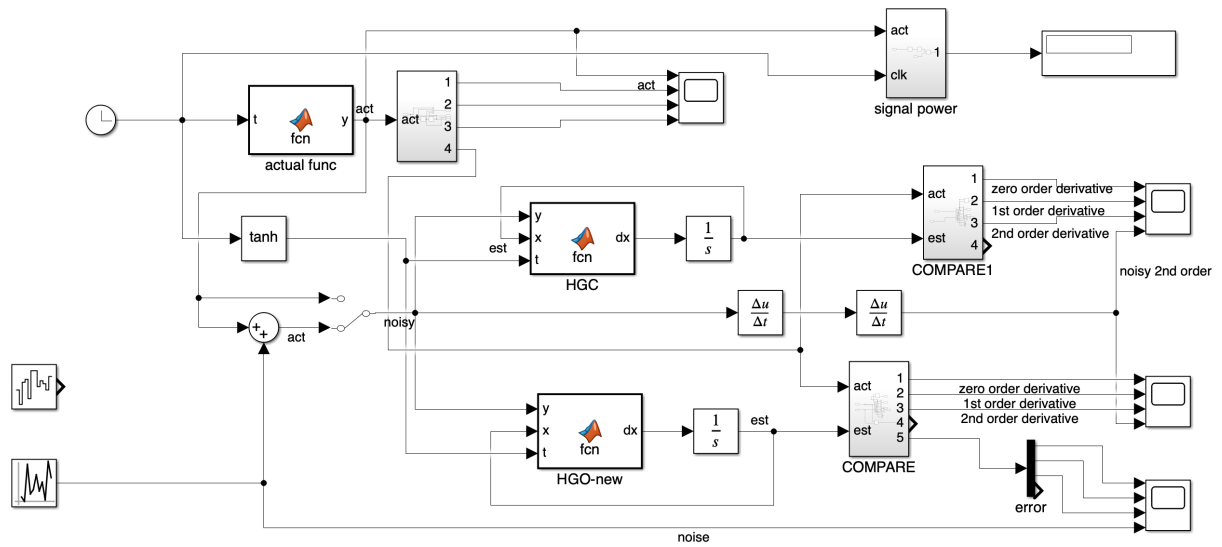


Figure 5.3: Modified HGO Simulink models

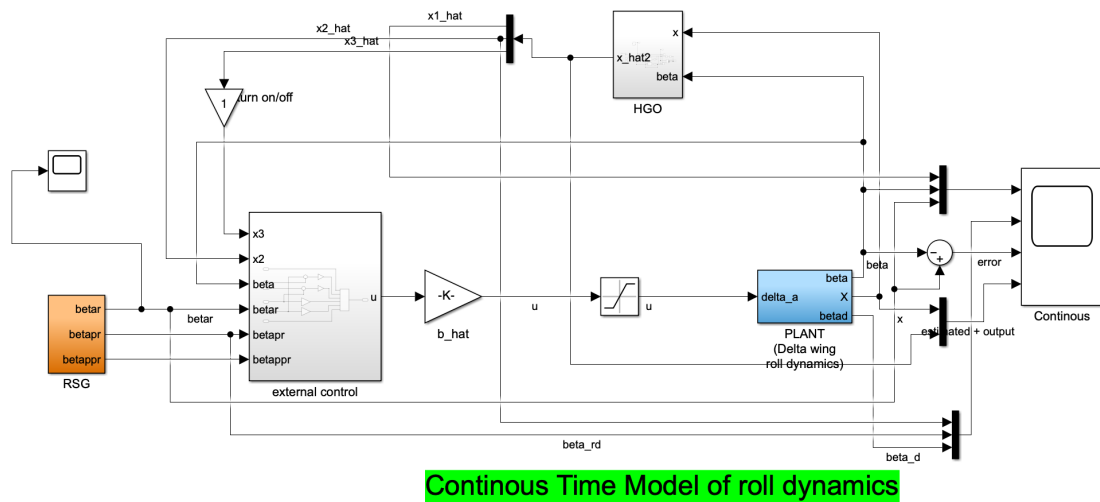


Figure 5.4: Model-free control with modified HGO Simulink models